Chinmayi Subramanya

subramanya@mpi-cbg.de
chinmayi-subramanya
Dresden, Germany

I am a PhD student in the IMPRS BioMAP program in Dr. Carl Modes' group at the Max Planck Institute of Molecular Cell Biology and Genetics.
I have a strong interdisciplinary foundation with background in theoretical physics, mathematics and biology.

Education

Max Planck Institute of Molecular Cell Biology and Genetics (MPI CBG) — **Dresden, Germany**
PhD in the IMPRS for Biology, Mathematics, AI, and Physics — 2025 - Present
Advisor: Dr. Carl Modes

Indian Institute of Science Education and Research (IISER) — **Mohali, India**
Integrated BS-MS in Physics — 2019 - 2024

Research Experience

Master's Thesis — **MPI CBG, Germany**
Title: Spectral Properties of Ensnarled Networks — September 2023 - May 2024
Advisor: Dr. Carl Modes

- Investigated the spectral properties of ensnarlment matrices which quantify intertwinedness between two spatially embedded vessel networks
- Derived a novel eigenvector centrality measure for the ensnarlment metric
- Applied the model to analyze the ensnarlment of bile canaliculi and sinusoidal networks in mouse liver sections using 3D imaging data

Summer Research Intern — **TU Dortmund, Germany**
Topic: S-Matrix elements of rare B-meson decays — June - July 2022
Advisor: Prof. Emmanuel Stamou

Summer Research Intern — **UC Irvine, USA**
Topic: Neutrino masses and modular forms — May - October 2021
Advisor: Prof. Mu-Chun Chen

Summer Research Intern — **IISER Mohali, India**
Topic: Modeling Rydberg States of Helium — June - August 2020
Advisor: Dr. P. Balanarayan

Manuscripts

C. Subramanya, Y. Tian and C. D. Modes, "Spectral Properties of Ensnarled Networks" *In preparation*

Y. Tian, **C. Subramanya** and C. D. Modes, "GEMINI: Generalized Ensnarlment Measure from Incomplete-linkage of Network-network Interactions" *Submitted*

Skills

- **Python:** Network simulations with networkx; data analysis and visualizations with numpy, pandas, scipy, pickle, matplotlib, seaborn
- **Software and tools:** Bash scripting, git vesion control system
- **Computation and visualization:** Wolfram Mathematica
- **Scientific writing:** LaTeX

Conferences and Workshops

Mechanobiology Across the Tree of Life — **EMBL Heidelberg**, June 2026
Poster: Ensnarled: Deciphering the architecture and physiology of complex interconnected networks
Best poster prize

Building Networks: Engineering in Vascular Biology — **EMBL Barcelona**, May 2026
Poster: Ensnarled: Deciphering the architecture and physiology of complex interconnected networks

ConFlowNet 2025 — **TU Munich, Germany**, September 2025
Poster: Spectral properties of ensnarled networks

2nd MPI Dresden-Leipzig Day — **MPI CBG**, June 2025
Short talk: Ensnarled: Deciphering the architecture of complex interconnected systems

Awards and Fellowships

IMPRS BioMAP Predoctoral Guest Fellowship — February - May 2025
Max Planck Institute of Molecular Cell Biology and Genetics
Fellowship for a wetlab training rotation in Dr. Meritxell Huch's lab

IMPRS BioMAP Thesis-Project Fellowship — September 2023 - March 2024
Max Planck Institute of Molecular Cell Biology and Genetics
Fellowship to pursue Master's thesis project with Dr. Carl Modes

INSPIRE Scholarship for Higher Education — 2019-2024
Department of Science and Technology, Government of India
High-school merit based award. Combined yearly scholarship and summer research grants of INR 4,00,000

Academic Excellence Certificates
IISER Mohali
Top grades of the class in semesters 3 and 4

Referees

Dr. Carl Modes
Reseach Group Leader
MPI-CBG and CSBD
Pfotenhauerstr. 108
01307 Dresden, Germany
modes@mpi-cbg.de
+49 351 210-1932

Prof. Mu-Chun Chen
Professor
Dept. of Physics and Astronomy
University of California
Irvine, CA 92697-4575, USA
muchunc@uci.edu
+1 (949) 385-0246

Dr. Abhishek Chaudhuri
Associate Professor
Dept. of Physical Sciences
IISER Mohali
Mohali, Punjab 140306, India
abhishek@iisermohali.ac.in
+91-172-224-0135